



Review Pulse v3.0 — Aspect-Based Sentiment Analysis

"One review, many opinions — sentiment for each aspect, with attention you can see."

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Approach · Contribution

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Intro · Literature

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Problem · Plan & Risks



THE PROBLEM

Why one label is not enough

"The food was great but the service was slow." One review, two opposite opinions — sentence-level models collapse it to a single neutral.



One review

two opposite opinions, one per aspect



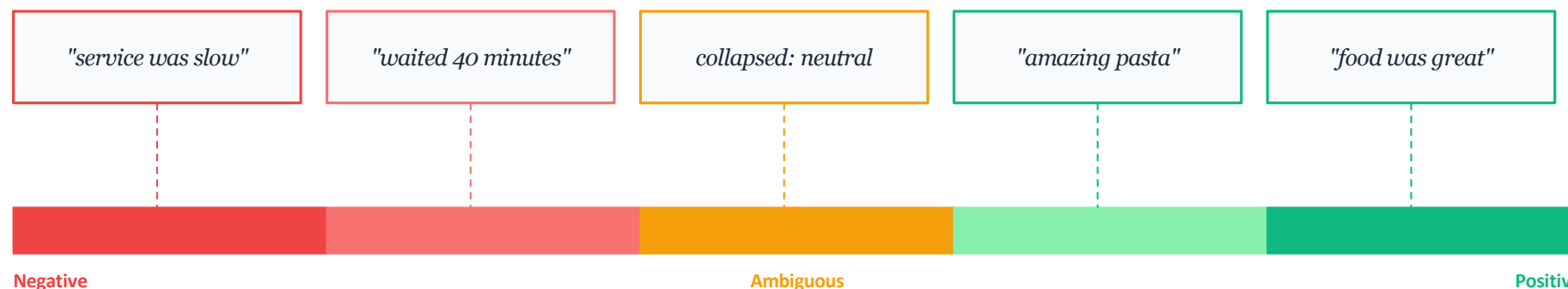
Sentence-level models

our A1 N-gram and Zhao et al. CNN both collapse to one label



Business value

"food good, service bad" is actionable — "neutral" is not

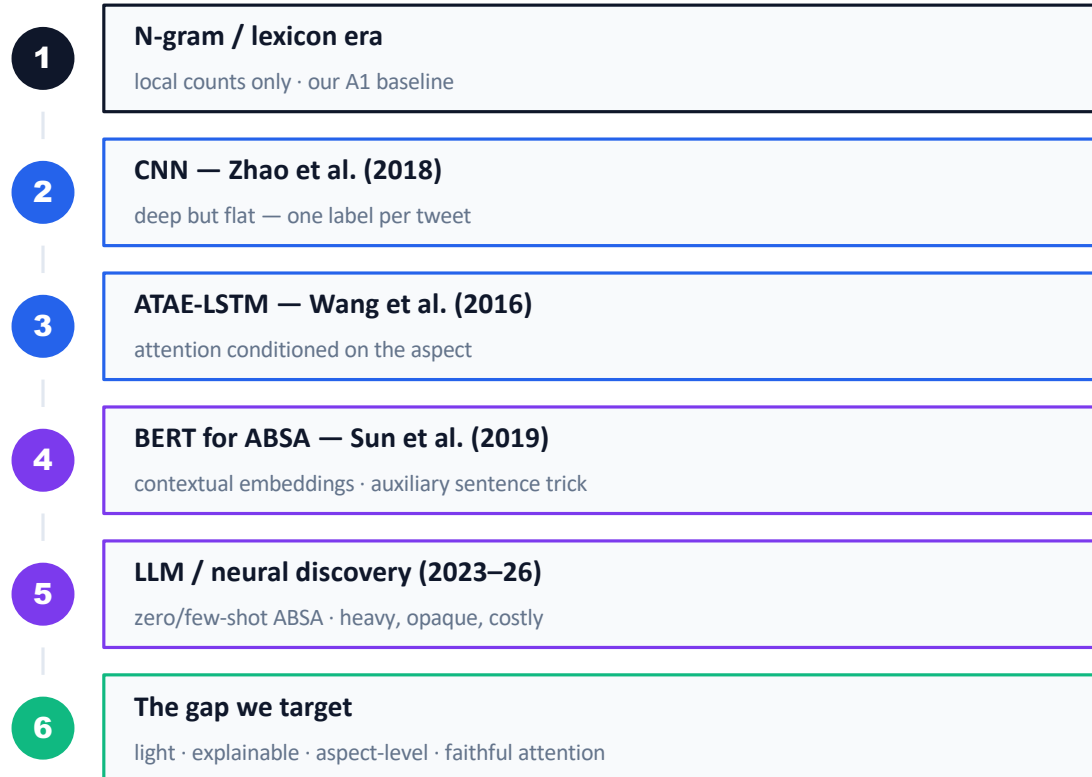


Sentence-level collapse — opposite opinions average into a meaningless neutral



WHAT THE LITERATURE SHOWS

One limitation — a single label per text — solved step by step



OUR APPROACH

Four-model ladder — interpretability is the differentiator

LADDER 1-2 · BASELINES

Sentence-level floor

```
Ladder(
  1. TF-IDF + LogReg
  2. LSTM no target
  GloVe 100d
)
Note(
  no attention · no heatmap
)
```

ROLE

Set the sentence-level floor

LIMIT

One label per review — no why

LADDER 3 · ATAE-LSTM

Aspect-aware attention

(review, aspect)
(review, aspect) pair input

Per-aspect sentiment
accuracy · macro-F1 · efficiency

Attention weights
faithful · inspectable

3-class output
neg · neutral · pos

ROLE

Explainable core — and the fallback if the transformer disappoints.

LADDER 4 · DISTILBERT

Fine-tuned transformer

Input (review, aspect)
pair input · AutoTokenizer · free Colab GPU

LSTM + attention
aspect-conditioned weights

Attention heatmap
which words drove each call

SCOPE

1. Restaurants (core) + Laptops (extension) — pre-annotated

2. Topic modelling for aspect discovery — gated stretch goal

Only ATAE-LSTM and DistilBERT expose attention — we show why, not just the label.



PLAN & RISKS

Modules 8–12: feasible and de-risked

Timeline: data + baseline → ATAE-LSTM → DistilBERT → interpretability + Streamlit demo → evaluation + report.



Small data → overfitting

RISK

Mitigated: dropout · early stopping · transfer learning from pretrained embeddings.



Heavy fine-tuning

RISK

Mitigated: DistilBERT fine-tuned on a free Colab GPU — zero-cost budget.



Scope creep

RISK

Mitigated: aspect extraction & topic modelling are gated stretch goals.



Pre-annotated data

WIN

SemEval-2014 ships with aspect labels — the feasibility win: no manual labelling.



Light compute

WIN

Free Colab · open datasets · open-source libraries — zero-cost end to end.



Working fallback

WIN

If the transformer underperforms, the ATAE-LSTM path still meets the rubric.



EXPECTED CONTRIBUTION

From one score per review to a transparent, per-aspect read

**PROPOSED** Review Pulse v3.0 — built and demoed live in Assessment 3**1**

ASPECT-LEVEL

Sentiment per thing

Per-aspect labels instead of one collapsed score per review.

2

EXPLAINABLE

Attention heatmap

Shows which words drove each per-aspect call — the why, not just the label. Our differentiator.

3

COMPARED

Four-model ladder

TF-IDF → LSTM → ATAE-LSTM → DistilBERT, evaluated on identical benchmarks — accuracy and efficiency.

4

LIVE DEMO

Streamlit app

Working demo in Assessment 3 — type a review, watch per-aspect sentiment and attention live.



Questions?

2

SemEval datasets

2

models compared

5

modules · 8 to 12

1

live demo in A3

References: Wang et al. (2016) · Sun et al. (2019) · Devlin et al. (2019) · He et al. (2017) · Pontiki et al. (2014) · Hua et al. (2024) · Zhao et al. (2018) — full APA list in the written proposal.

[GitHub](#) github.com/lfariabr/review-pulse · ReviewPulse v3.0 · DLE602 Assessment 2

